

ANNA UNIVERSITY, CHENNAI – 600 044
MADRAS INSTITUTE OF TECHNOLOGY CAMPUS

DEPARTMENT OF COMPUTER TECHNOLOGY

II YEAR
ACADEMIC YEAR 2026 – 2027

A Project Report on

INFOVERSE

A Digital Knowledge Encyclopedia

Submitted by
NAINA BAWANTHADE
Register No: 2025503070

Under the Guidance of
Dr. Ponsy R.K. Sathia Bhama

2025503070

CERTIFICATE

This is to certify that the project report entitled **"INFOVERSE – A DIGITAL KNOWLEDGE ENCYCLOPEDIA"** is the bonafide work of **Naina Bawanthade** (Register No: 2025503070), II Year, Department of Computer Technology, who carried out the project work under my supervision during the academic year 2026–2027.

Submitted for the project viva-voce examination held on _____.

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Signature of Guide	Signature of Head of Department
Dr. Ponsy R.K. Sathia Bhama	Dr. Ponsy R.K. Sathia Bhama

DECLARATION

I, **Naina Bawanthade** (Register No: 2025503070), hereby declare that the project report titled **"INFOVERSE – A DIGITAL KNOWLEDGE ENCYCLOPEDIA"** is an original record of the work carried out by me under the guidance of **Dr. Ponsy R.K. Sathia Bhama,** Department of Computer Technology, and that this work has not been submitted, either in part or in full, for the award of any other degree or diploma of this or any other institution.

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Place: Chennai

Date: September 2026

Signature of the Student

Naina Bawanthade
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ACKNOWLEDGEMENT

I express my sincere gratitude to my project guide, Dr. Ponsy R.K. Sathia Bhama, Department of Computer Technology, for her valuable guidance, constant encouragement and support throughout the course of this project. Her insightful suggestions were instrumental in shaping the design and implementation of Infoverse.

I am also thankful to the Head of the Department and all the faculty members of the Department of Computer Technology, Madras Institute of Technology Campus, Anna University, for providing the necessary facilities and a conducive environment to carry out this project.

I would like to extend my thanks to my family and friends for their constant motivation and support during the development of this project.

Finally, I thank the Almighty for giving me the strength and perseverance to complete this project successfully.

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ABSTRACT

Infoverse is a web-based digital knowledge encyclopedia designed to provide users with a centralized, organized and easy-to-navigate platform for accessing information across multiple domains, including Technology, Science, Education, History, Sports and Health. In an age where information is scattered across countless unreliable sources, Infoverse aims to bring together well-structured content in a single, user-friendly interface, similar in concept to popular knowledge platforms such as Wikipedia, but built and maintained at a scale suitable for an academic project.

The current version of Infoverse is developed using HTML, CSS and JavaScript, and offers a responsive design that adapts smoothly to desktop, tablet and mobile screens. Core features already implemented include a functional search bar, category-based browsing, an About Project section, a Project Gallery, a Contact section and smooth client-side interactivity such as animations and a back-to-top button.

This project report documents the restructuring of Infoverse into five well-defined modules — Search & Information Retrieval, Category Management, Information Display, Content Management, and User & Admin Management — developed using the Agile software development model. The report also covers the problem statement, objectives, proposed system, system architecture, project scheduling (Gantt chart, AON and AOE diagrams), technologies used, testing methodology and future enhancements planned for Infoverse.

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INTRODUCTION

The internet has become the primary source of information for students, researchers and the general public. However, information relevant to a specific topic is often scattered across multiple websites, many of which are cluttered with advertisements, inconsistent formatting or unreliable content. This creates difficulty for users who want quick, organized and trustworthy access to information.

Infoverse addresses this problem by acting as a centralized "digital knowledge encyclopedia" — a single website where information from various domains is organized under clear categories and made easily searchable. The platform currently supports six major categories: Technology, Science, Education, History, Sports and Health, and is built with a strong focus on simplicity, responsiveness and usability.

This project was undertaken as part of the academic curriculum to apply concepts of web development, software engineering and project management in building a real, functioning system. Rather than building an entirely new platform, this project focuses on improving and restructuring the existing Infoverse website into clearly defined modules, while retaining its original design philosophy and purpose.

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PROBLEM STATEMENT

Users looking for general information on topics such as science, history, technology or health often face the following difficulties:

- Information is scattered across many different websites with inconsistent quality and formatting.
- Existing general-purpose search engines return excessive and sometimes irrelevant results, requiring users to filter through multiple sources.
- Many information websites are not optimized for mobile or tablet devices, resulting in a poor user experience.
- There is no single, lightweight, easy-to-navigate platform that organizes information into clear, browsable categories for quick access.
- Content management on many small-scale information platforms is not modular, making it difficult to add, edit or remove information efficiently.

Infoverse aims to solve these problems by providing a centralized, responsive and well-organized platform where users can search for and browse information by category, while administrators can manage content through a dedicated, modular content management system.

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OBJECTIVES

The primary objectives of the Infoverse project are as follows:

- To design and maintain a centralized web platform for accessing organized information across multiple categories.
- To provide an efficient search feature that allows users to quickly locate relevant topics.
- To organize content into well-defined categories: Technology, Science, Education, History, Sports and Health.
- To ensure the website is fully responsive and accessible across desktop, tablet and mobile devices.
- To restructure the existing website into five clear, manageable modules for easier development, testing and maintenance.
- To apply the Agile development model, allowing each module to be built and improved iteratively.
- To provide a simple and secure mechanism for administrators to add, edit and delete content.
- To document the system using standard project management tools such as Gantt charts and AON/AOE network diagrams.

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PROPOSED SYSTEM

The proposed system builds upon the existing Infoverse website rather than replacing it. The current site already includes a home page, search bar, category pages, an About Project section, a Project Gallery, a Contact section and several supporting sections such as SDLC, Gantt Chart and AON/AOE. The proposed enhancement focuses on reorganizing the underlying functionality of the website into five clearly defined, independently manageable modules.

Under the proposed system, each functional area of Infoverse — searching, category browsing, information display, content management and user/admin control — is treated as a separate module with its own defined responsibility. This modular restructuring makes the system easier to understand, test and extend in future iterations, without requiring a complete redesign of the existing HTML, CSS and JavaScript codebase.

Key improvements proposed for the current version of Infoverse include:

- Cleaner separation between the search logic, category logic and content display logic in the JavaScript code.
- A more structured way of storing and retrieving category-wise content.
- A simplified admin-facing workflow for adding, editing and deleting content within categories.
- Improved navigation between categories, search results and individual topic pages.
- Continued responsive design so that all new functionality works correctly on desktop, tablet and mobile screens.

The system continues to be a purely front-end, client-side project (HTML, CSS, JavaScript), keeping the scope realistic and appropriate for a second-year academic project, while still demonstrating sound principles of modular software design.

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MODULES

To make development, testing and maintenance easier, the Infoverse project is organized into the following five modules. Each module is developed and refined iteratively using the Agile model.

1. Search & Information Retrieval Module

This module allows users to search for topics using the search bar on the home page. It matches the user's query against titles and keywords of the available content and displays the most relevant results. This module also handles smooth navigation to the corresponding information page once a result is selected.

2. Category Management Module

This module organizes all information on the website into six main categories: Technology, Science, Education, History, Sports and Health. It manages how topics are grouped, listed and displayed under each category page, allowing users to browse information without needing to search directly.

3. Information Display Module

This module is responsible for presenting the detailed content of a selected topic or category to the user in a clean, readable format. It handles the layout of text, headings and any supporting images associated with a topic, ensuring a consistent reading experience across the site.

4. Content Management Module

This module handles the addition, editing and deletion of information within the platform. It is primarily used by the site administrator to keep the content of each category up to date, and provides a simple interface for managing topics without directly editing the underlying code.

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5. User & Admin Management Module

This module manages access control for the website, distinguishing between regular users (who can search and browse information) and the administrator (who has additional access to the Content Management Module). It ensures that content modification is restricted to authorized access only.

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AGILE MODEL / SDLC

The Infoverse project follows the Agile software development model. Unlike the traditional Waterfall model, where each phase must be completed before moving to the next, the Agile model allows the project to be broken down into small, manageable iterations. Each of the five modules of Infoverse is treated as a mini development cycle, allowing continuous improvement based on testing and feedback.

Each iteration in the Agile model followed for this project generally consists of the following phases:

- Planning — Identifying the requirements and scope of the module to be developed.
- Design — Designing the layout, structure and logic required for the module.
- Development — Implementing the module using HTML, CSS and JavaScript.
- Testing — Verifying that the module works correctly and integrates well with existing modules.
- Review — Evaluating the module's performance and identifying improvements for the next iteration.

This iterative approach was chosen because it allows individual modules — such as Search & Information Retrieval or Content Management — to be developed and refined independently, while still contributing to the overall functioning website. It also makes it easier to accommodate changes or improvements discovered during development, which is especially useful for a project undergoing continuous enhancement like Infoverse.

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SYSTEM ARCHITECTURE

The system architecture of Infoverse follows a layered structure, separating the presentation, application and data-handling responsibilities of the website. This separation makes the five modules easier to develop, test and maintain independently.

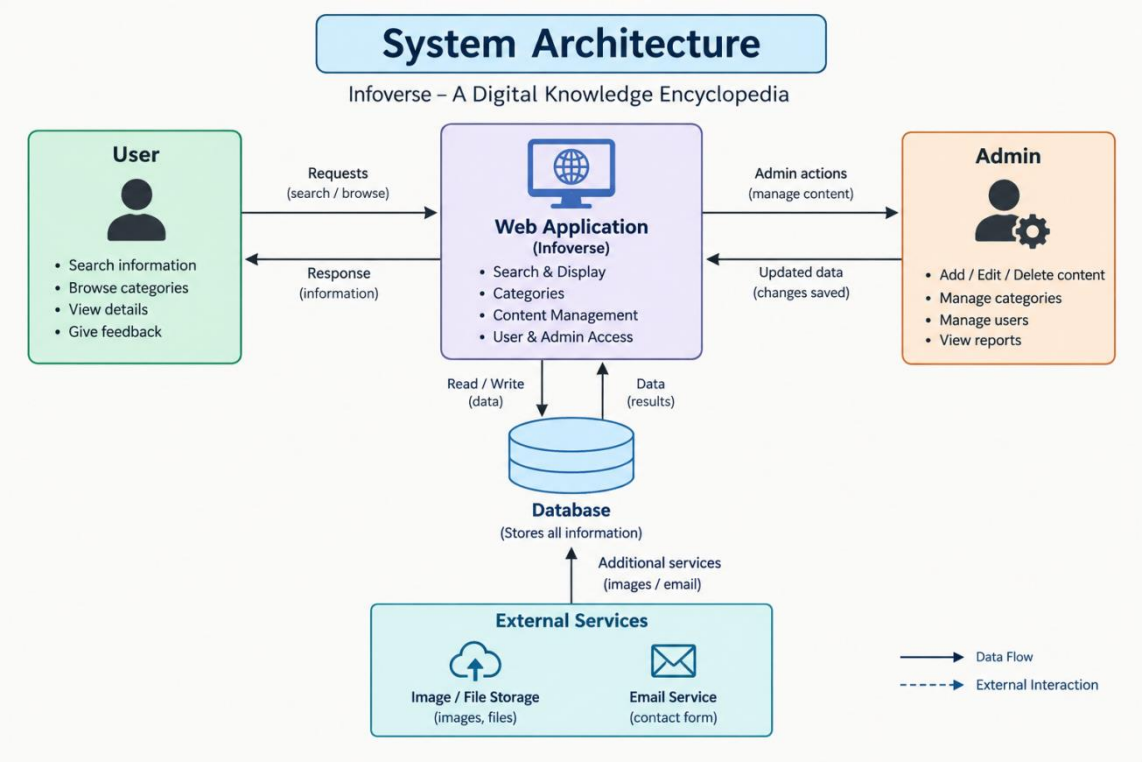


Fig. 1: System Architecture of Infoverse

At the top level, users and administrators interact with Infoverse through a standard web browser. The Presentation Layer, built using HTML, CSS and JavaScript, renders the user interface and handles client-side interactivity such as animations, smooth navigation and the back-to-top button. The Application Layer contains the logic for all five modules, coordinating search operations, category organization, content display and administrative actions. Finally, the underlying category and content data is organized and accessed by the relevant modules, with the Admin Panel providing controlled access for content management.

MODULE FLOW DIAGRAM

The module flow diagram below illustrates the typical flow of control when a user interacts with Infoverse — from entering a search query to viewing the requested information, along with the supporting role of the Content Management and User & Admin Management modules.

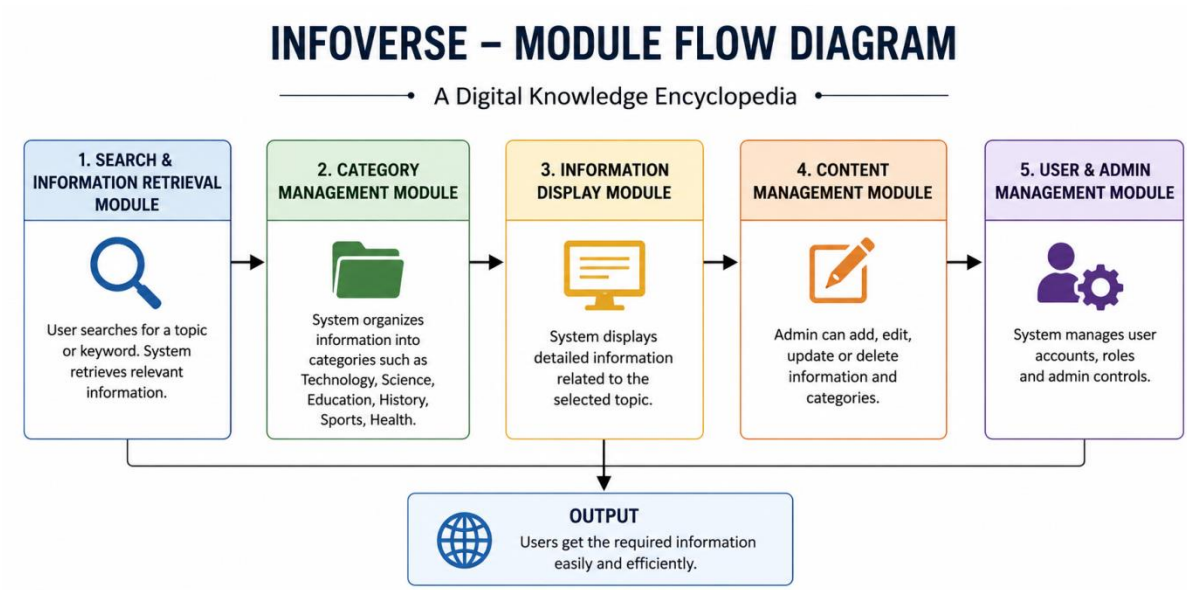


Fig. 2: Module Flow Diagram of Infoverse

When a user enters a search query, the Search & Information Retrieval Module processes the request and works together with the Category Management Module to locate matching topics. The relevant content is then rendered to the user through the Information Display Module. The Content Management Module operates in the background, allowing the administrator to keep information current, while the User & Admin Management Module governs access rights across the system.

TECHNOLOGIES USED

Infoverse is developed as a client-side web application. The following table summarizes the core technologies used in building and maintaining the platform.

Technology	Purpose
HTML5	Provides the basic structure and content of all web pages.
CSS3	Handles styling, layout and responsive design across desktop, tablet and mobile screens.
JavaScript	Implements interactivity: search functionality, smooth navigation, contact-form handling, animations and the back-to-top button.
Responsive Design (Media Queries)	Ensures the website adapts correctly to different screen sizes and devices.
Git & GitHub	Used for version control and maintaining the project source code.
Visual Studio Code	Code editor used for developing and testing the website.

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GANTT CHART

The Gantt chart below represents the planned schedule for the development of Infoverse across a 12-week timeline, showing the approximate duration allotted to each phase and module.

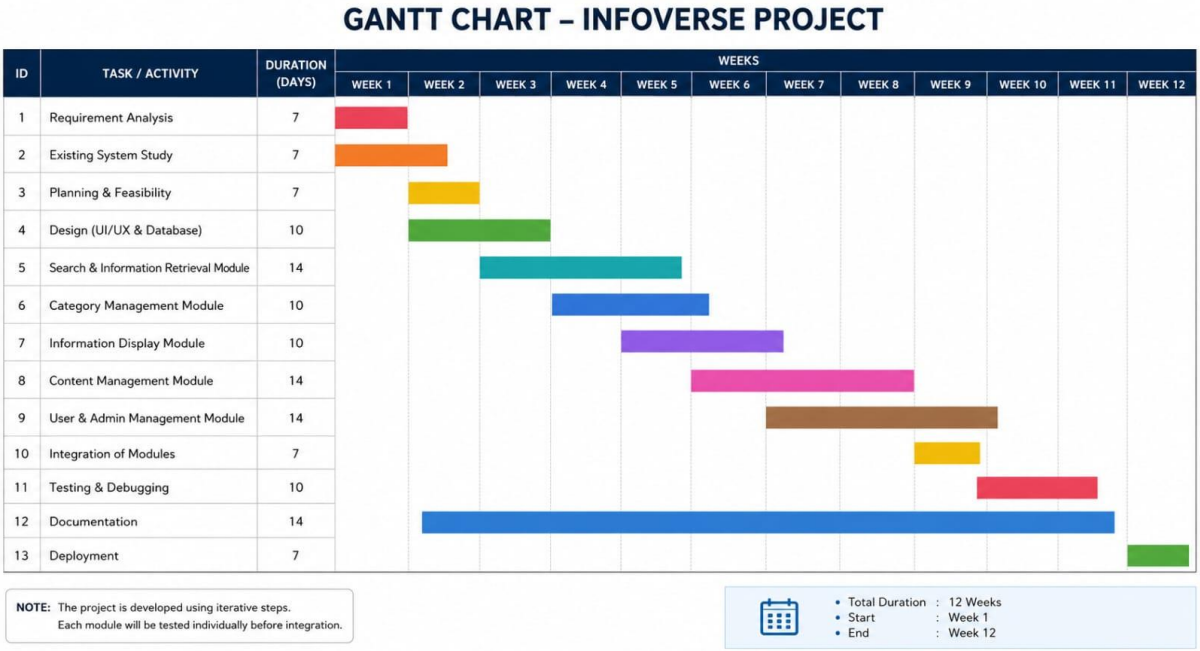


Fig. 3: Gantt Chart for Infoverse Project Schedule

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AON/AOE DIAGRAM

Activity on Node (AON) Diagram

The AON diagram represents each activity of the project as a node, with arrows indicating the sequence and dependency between activities. It shows that requirement analysis and design must be completed before the five modules are developed, and that all modules must be completed and integrated before final testing.

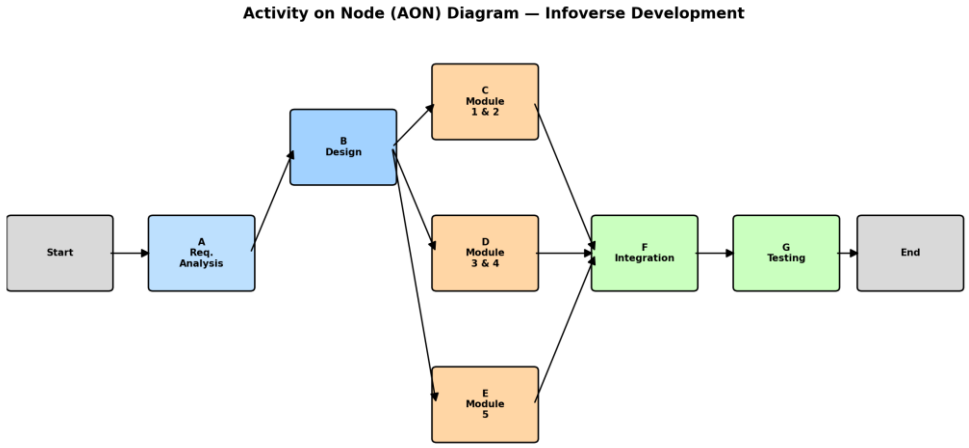


Fig. 4: Activity on Node (AON) Diagram

Activity on Edge (AOE) Diagram

The AOE diagram represents each activity as a directed edge between two events (numbered circles), with the estimated duration of each activity labelled along the edge. This helps identify the critical path — the sequence of activities that determines the minimum time required to complete the project.

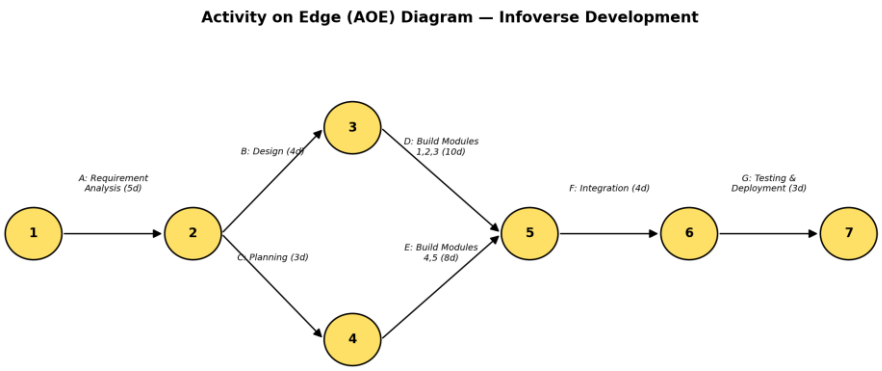
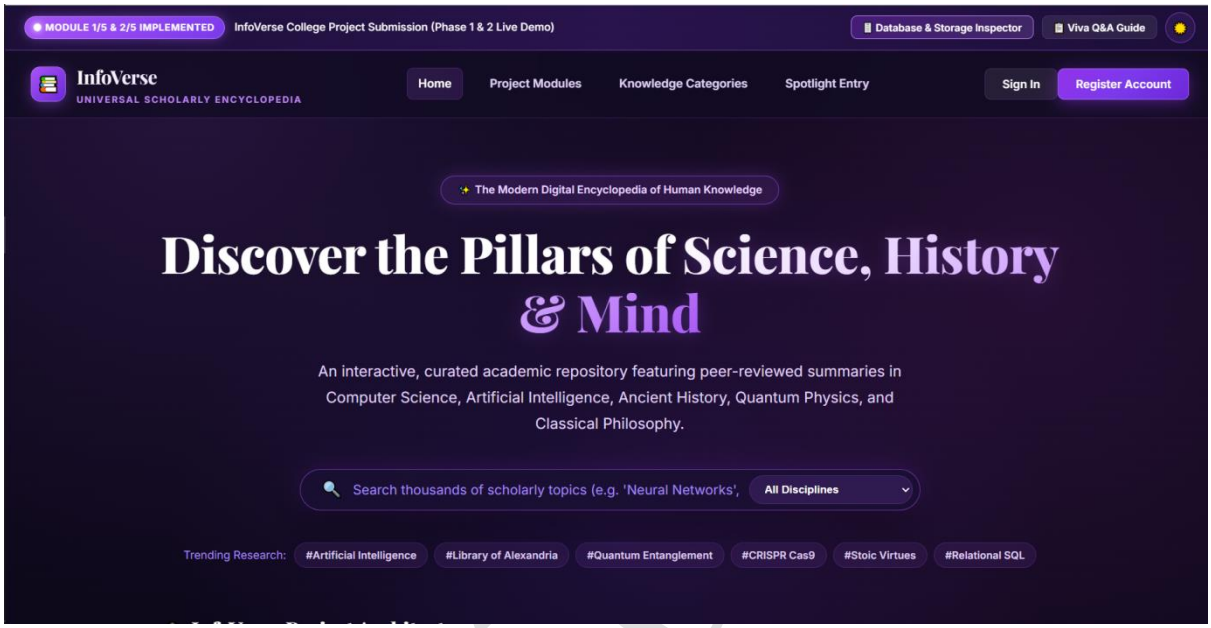


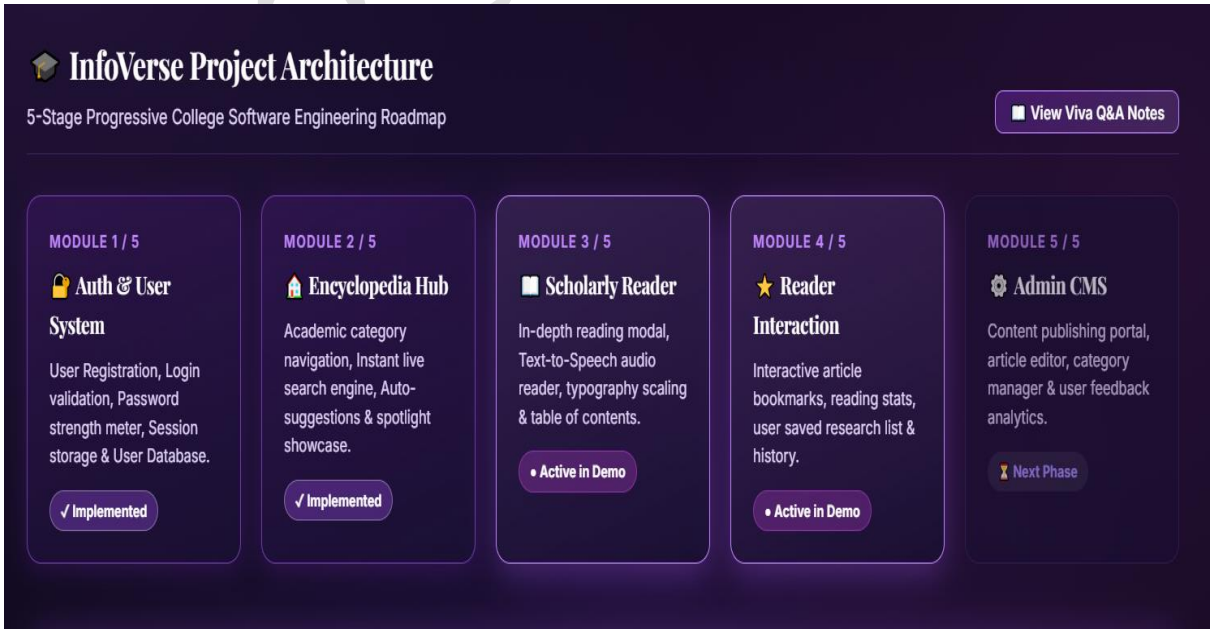
Fig. 5: Activity on Edge (AOE) Diagram

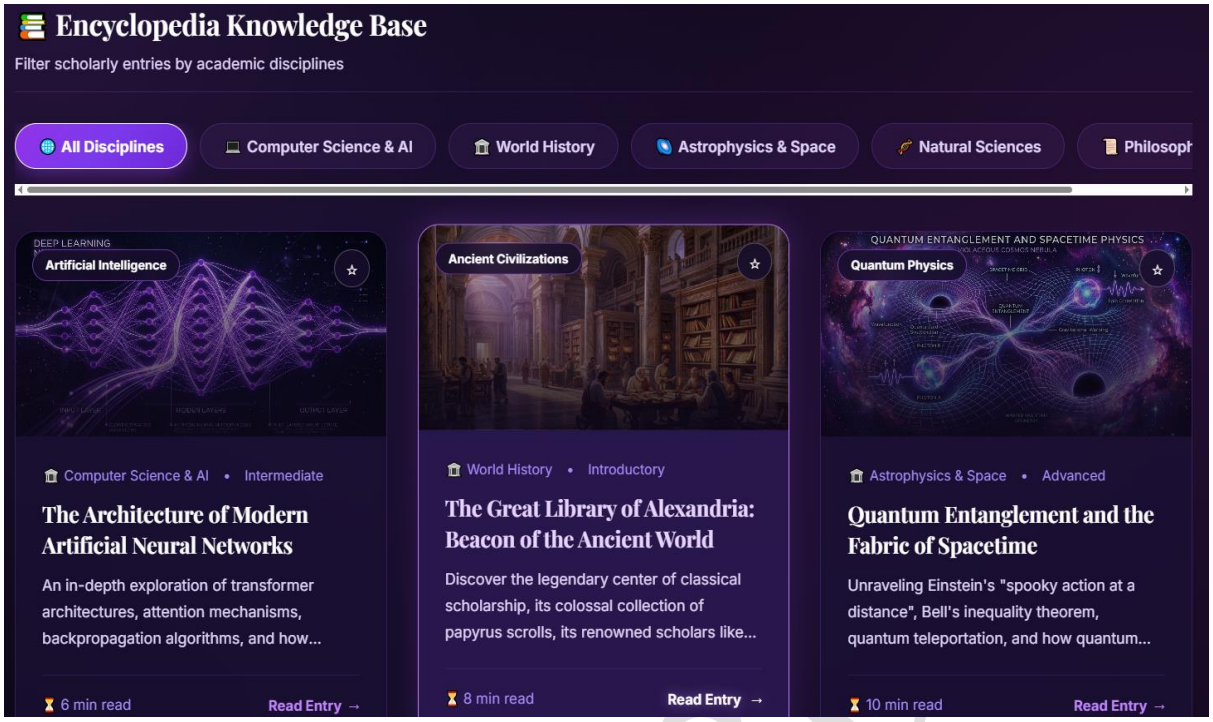
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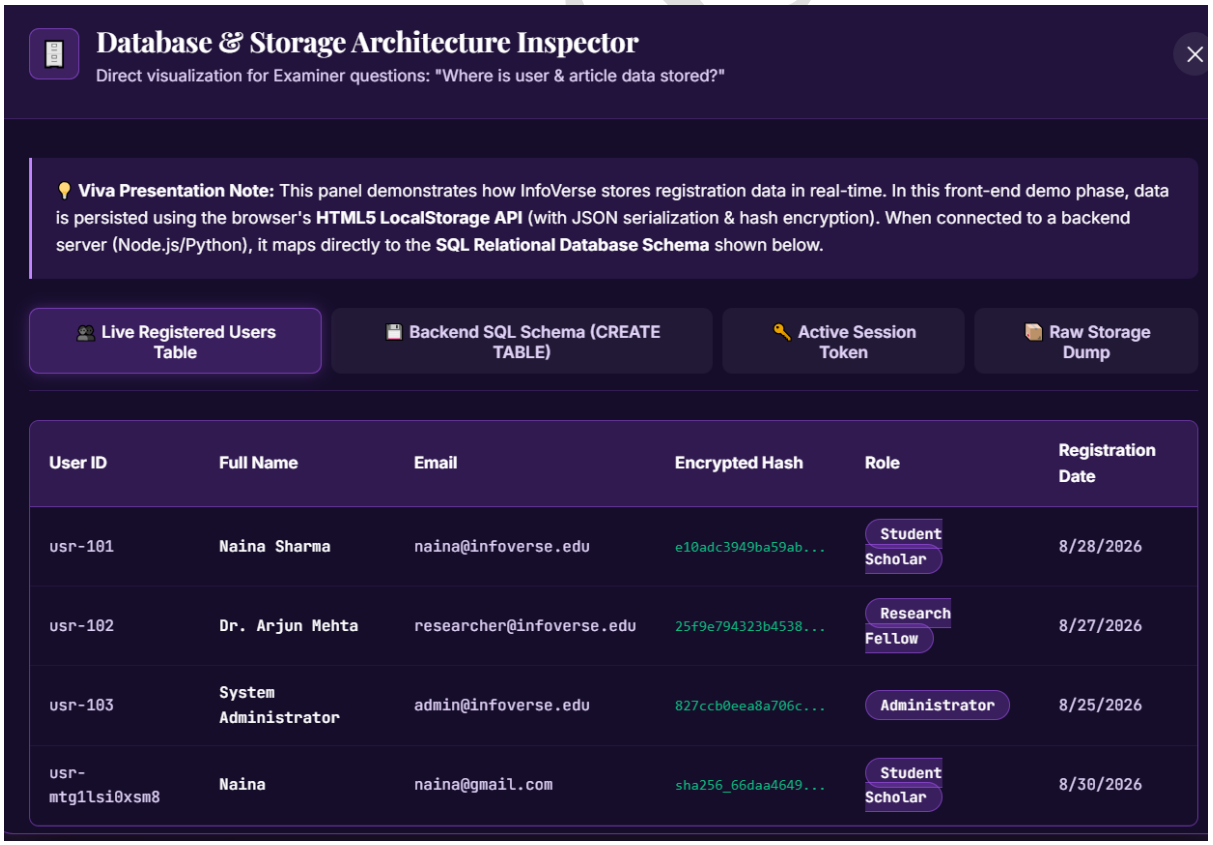
SCREENSHOTS

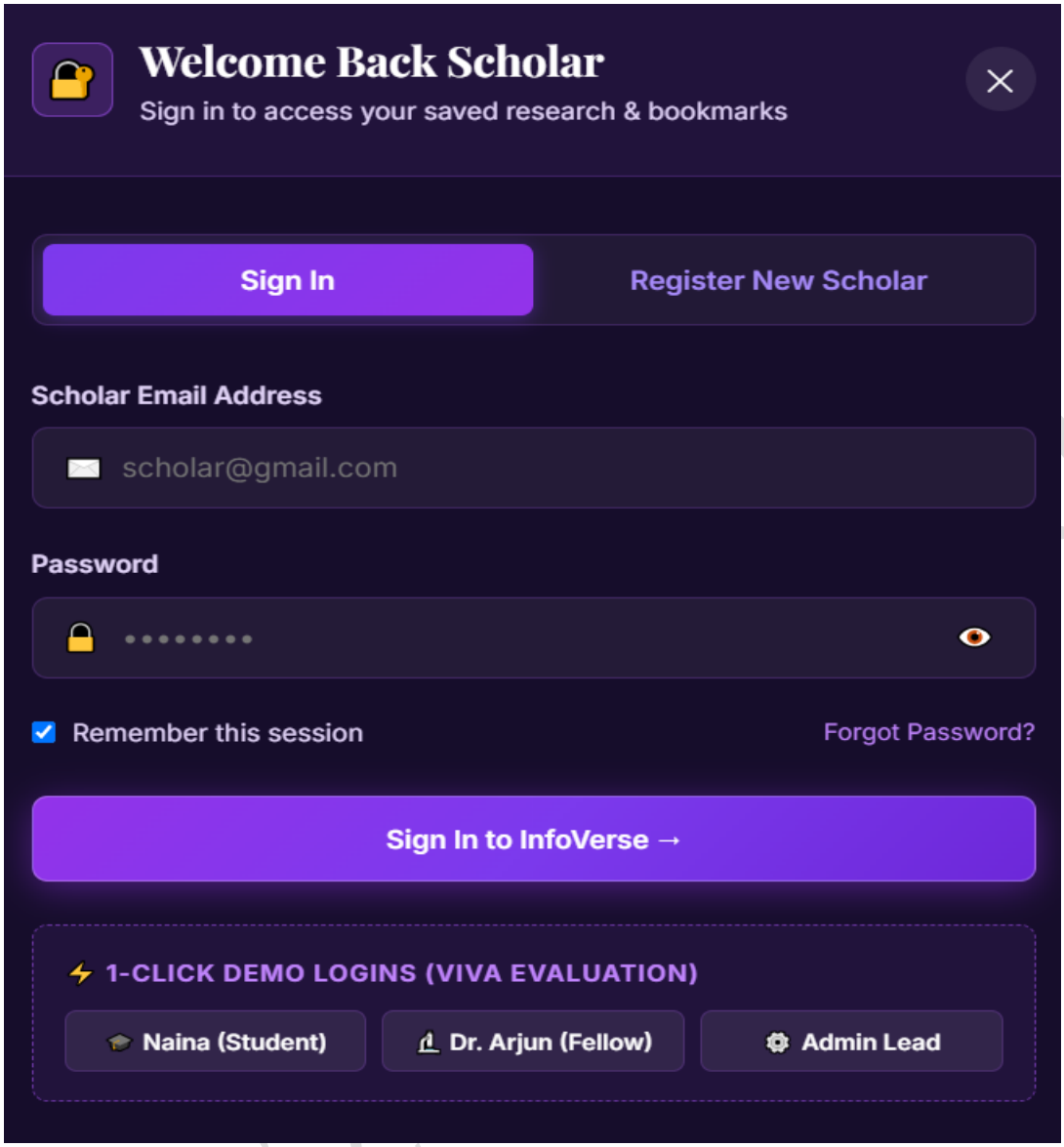
This section is reserved for screenshots of the working Infoverse website, including the Home Page, Search Results, Category Pages, Information Display Page, Admin/Content Management interface and the Contact section.

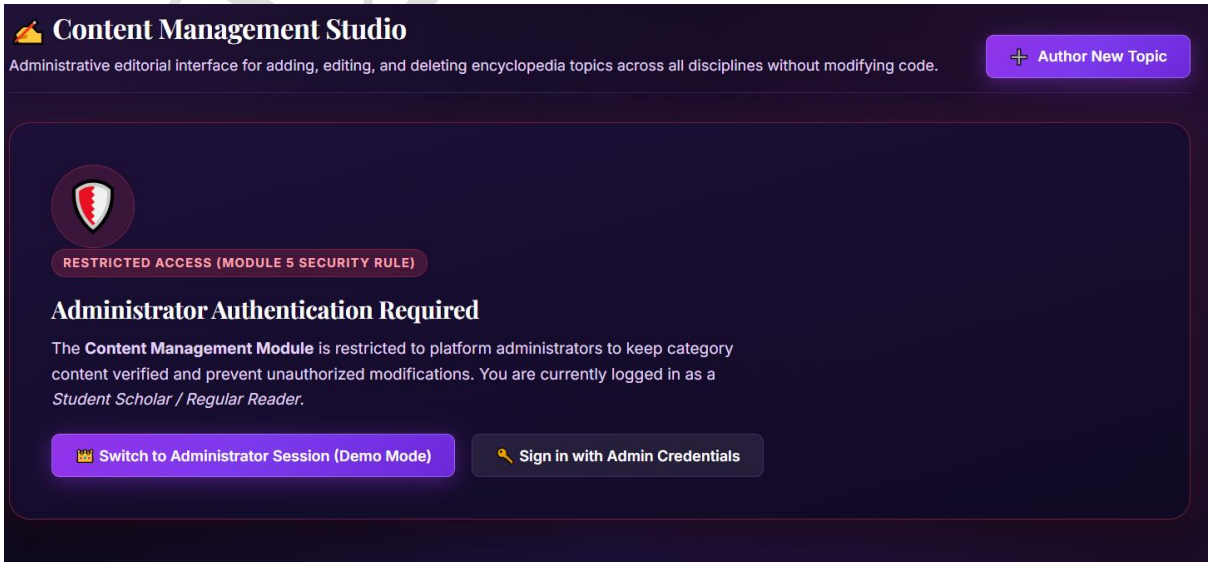





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Database & Storage Architecture Inspector
Direct visualization for Examiner questions: "Where is user & article data stored?"
Viva Presentation Note: This panel demonstrates how InfoVerse stores registration data in real-time. In this front-end demo phase, data is persisted using the browser's HTML5 LocalStorage API (with JSON serialization & hash encryption). When connected to a backend server (Node.js/Python), it maps directly to the SQL Relational Database Schema shown below.
Live Registered Users Table
Backend SQL Schema (CREATE TABLE)
Active Session Token
Raw Storage Dump
User ID
Full Name
Email
Encrypted Hash
Role
Registration Date
usr-101
Naina Sharma
naina@infoverse.edu
e10adc3949ba59ab...
Student Scholar
8/28/2026
usr-102
Dr. Arjun Mehta
researcher@infoverse.edu
25f9e794323b4538...
Research Fellow
8/27/2026
usr-103
System Administrator
admin@infoverse.edu
827ccb0eea8a706c...
Administrator
8/25/2026
usr-mtg1lsi0xsm8
Naina
naina@gmail.com
sha256_66daa4649...
Student Scholar
8/30/2026


Welcome Back Scholar
Sign in to access your saved research & bookmarks
Sign In
Register New Scholar
Scholar Email Address
scholar@gmail.com
Password
Remember this session
Forgot Password?
Sign In to InfoVerse →
1-CLICK DEMO LOGINS (VIVA EVALUATION)
Naina (Student)
Dr. Arjun (Fellow)
Admin Lead


Content Management Studio
Administrative editorial interface for adding, editing, and deleting encyclopedia topics across all disciplines without modifying code.
Author New Topic
RESTRICTED ACCESS (MODULE 5 SECURITY RULE)
Administrator Authentication Required
The Content Management Module is restricted to platform administrators to keep category content verified and prevent unauthorized modifications. You are currently logged in as a Student Scholar / Regular Reader.
Switch to Administrator Session (Demo Mode)
Sign in with Admin Credentials

TESTING

Testing for Infoverse was carried out module-wise, in line with the Agile development approach, so that each module was verified independently before being integrated with the rest of the system. Manual functional testing was performed across desktop, tablet and mobile screen sizes to ensure consistent behaviour.

S.No	Test Case	Expected Result	Actual Result
1	Search for a valid topic keyword	Relevant topic results are displayed	Pass
2	Search for a keyword that does not exist	A 'No results found' message is shown	Pass
3	Click on a category (e.g., Science)	All topics under that category are listed	Pass
4	Select a topic from search results	Detailed information page loads correctly	Pass
5	Admin adds a new topic under a category	New topic is saved and visible under the category	Pass
6	Admin edits/deletes an existing topic	Changes are reflected correctly on the site	Pass
7	Submit the contact form with valid details	Confirmation message is displayed	Pass
8	Open the website on a mobile screen	Layout adjusts responsively without breaking	Pass

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FUTURE ENHANCEMENTS

While the current version of Infoverse is fully functional as a front-end, client-side platform, several enhancements can be considered for future iterations of the project:

- Integrating a backend database (such as MySQL or MongoDB) to store content dynamically instead of static files.
- Adding user accounts, allowing users to bookmark or save favourite topics.
- Implementing a more advanced search algorithm with auto-suggestions and filters by category.
- Introducing a rich-text editor in the Content Management Module for easier formatting of new content.
- Adding a dark mode / theme-switching option for improved accessibility.
- Allowing registered users to contribute or suggest edits to existing topics, subject to admin approval.
- Expanding the category list beyond the current six categories based on user demand.

GitHub Link

https://github.com/Tiara200718/INFOVERSE---A-DIGITAL-ENCYLOPEDIA

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CONCLUSION

Infoverse successfully demonstrates the design and development of a centralized, responsive web platform for organizing and accessing information across multiple domains. By restructuring the existing website into five clearly defined modules — Search & Information Retrieval, Category Management, Information Display, Content Management, and User & Admin Management — the project achieves a modular, maintainable structure while preserving the original design and purpose of the site.

The application of the Agile development model allowed each module to be developed, tested and improved iteratively, resulting in a stable and functional website. Overall, this project provided valuable hands-on experience in front-end web development, modular software design and basic project management practices such as scheduling with Gantt charts and AON/AOE network diagrams.

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